



# Precision loss modeling method of ball screw pair

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## ABSTRACT

A precision loss modeling method of the ball screw pair was proposed by introducing an effective contact coefficient to consider the intermittent contact between the balls and screw grooves and the parameter in the precision loss model was modified by establishing a multiscale contact mechanics model. The multiscale contact mechanics model between two rough curved surfaces was established by considering the elastic, elastic-plastic, and plastic deformations of contacting asperities to compute such contact parameters as the total actual contact area and contact load. A surface contact coefficient was introduced into the multiscale contact mechanics model to modify the contact parameters by taking the rough curved surfaces of balls and screw grooves into account. The microtopographies of rough surfaces of screw grooves and balls were characterized by the fractal geometry theory, and then the characterization of the microtopography of screw groove surfaces and the multiscale contact mechanics model were unified in the precision loss modeling. Then the relative sliding distance was calculated according to the kinematic analysis of the ball screw. Finally, the precision loss model of the ball screw pair was established by the above analysis. To verify the effectiveness of the precision loss model, the wear test was conducted on a dual-drive ball screw feed drive system under different working conditions. The predicted results agree well with the experimental data in the stable wear stage and the predictive accuracy of the model is much higher than that of previous studies. The effects of such parameters as the axial load and the feed rate on the precision loss were discussed. It is shown that the precision loss increases with the axial load and the feed rate.

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## 1. Introduction

The urgent demand of modern manufacturing is the high-speed, high-accuracy, and heavy-load machining. The wide application of the ball screw feed drive system (BSFDS) in CNC machine tools is essential for the realization of the high-speed, high-accuracy, and heavy-load machining because the BSFDS is one of the key functional components of CNC machine tools to transfer rotational movement into linear movement with low friction torque. The positioning accuracy of BSFDSs, which is mainly affected by the control system error, the thermally-induced error, and the wear between screw grooves and rolling elements, is crucial for the machining accuracy of CNC machine tools. To improve the positioning accuracy, both the optimal control method [1,2] and the thermal error control technique [3–8] are used to reduce the control error and thermally-induced error, respectively. Then the effect of the wear on positioning accuracy is becoming more and more prominent. Especially, the wear between the ball screw groove and balls are accelerated by the high-speed and heavy-

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load working conditions, and then the service life, the service performance, and the positioning accuracy of the BSFDS is reduced significantly. So the positioning error caused by the wear of screw grooves is much more obvious than before. The wear issue between screw grooves and rolling elements is complicated by the coupling among such factors as the microtopography of rough surfaces, the feed rate, the running time, the material properties, and the external load. It is concluded that the investigation of the precision loss modeling method of the screw groove is beneficial for the improvement of the service life, the service performance, and the positioning accuracy of BSFDSs.

The wear mechanism between two rough surfaces has been extensively investigated and a large number of generalized models are established so far. The Archard wear model is the basic theoretical model to predict the wear volume between two rough surfaces. A fractal geometry model was proposed to predict the wear rate in terms of two fractal parameters for wear prediction and the results showed that the relationship between the wear rate  $V$  and the true contact area  $A_0$  is  $V \propto A_0^{m(D)}$ , wherein  $m(D)$  is a function of fractal dimension  $D$  and has a value between 0.5 and 1 [9,10]. A new two-body abrasion model was presented by combining the kinematic theory describing the variations in the axial wear depths with the operating strokes [11]. Brecher et al. [12] investigated the effect the oil additives on the wear behavior and preload loss in ball screws for machine tool applications and the results showed that the oil additives had a significant influence on the preloading of the ball screws. A generalized adhesive wear model was proposed to consider the effect of the interfacial adhesion on the total load for three-dimensional fractal surfaces in normal contact [13]. The current state and the development in the application of fractal geometry to the characterization of rough surfaces, the elastic and plastic contacts, the wear modeling and prediction, the frictional temperature distribution of rough surfaces, and the debris analysis were reviewed [14]. The theoretical relationship between the real contact area and the fractal parameters, the material properties, and the external load, etc. was obtained. Using the relationship and Rowe's modified adhesion theory, a fractal model of the running-in wear was developed [15]. A fatigue wear model of the plane sliding pairs under dry friction was proposed by establishing a dynamic contact model of surfaces and developing a mean fatigue damage constant of asperities [16]. All the above general wear models advance the development in the wear field and they are applicable to predict the wear between two rough surfaces. But the wear between balls and grooves in the BSFDS is the wear between two rough curved surfaces. For the wear between the balls and grooves, the surface mode changes from the rough surfaces to rough curved surfaces, and then the actual contact area and contact load, which participate in the wear process, are changed. The change in the above contact parameters leads to the variation of the wear volume. It is seen that the wear between two rough curved surfaces is totally different from that between two rough surfaces because the actual contact area and contact load which participate in the wear process are different. So the above models are not applicable to predict the wear between balls and screw grooves.

The wear issues in the rolling functional components were also investigated by researchers. A quasi-dynamic method was proposed to evaluate the wear characteristics of ball bearings and the permissible wear extent was confirmed based on the Archard wear equation [17]. The wear process of roller linear guides was analyzed based on the Archard wear theory and then a model to compute the displacement of the slider was proposed [18]. The precision loss of the ball screw was investigated by considering the full ball load distribution and the groove wear model was built to predict the precision loss of ball screws at different structural parameters and operating parameters [19]. The wear volume of ball-screw contacts was computed by the modified Archard theory and then the axial precision loss was calculated [20]. So far, the wear mechanism between two rough curved surfaces has not been fully investigated and a new model to predict the wear volume and precision loss of the ball screw pair should be proposed. The rough surfaces of grooves and balls are made up of a large number of asperities with different scales and then the elastic, elastic-plastic, and plastic deformations happen to the grooves and balls when the axial and radial loads and bending moments are applied to BSFDSs. The rough surfaces have the disorder, random, self-affinity, fractal, and multiscale behaviors, resulting in the fractal and multiscale characteristics of microcontacts. To compute the wear volume and precision loss of the screw groove, the microtopography of rough surfaces of the screw groove and the rolling element should be characterized uniquely because the microtopography directly determines such contact parameters as the actual contact area and contact load, and these contact parameters determine the wear volume directly. However, the above wear models assumed that the rough surfaces of the screw groove and the rolling element are perfectly smooth, resulting in the ignorance of the imperfect contact between them. The actual contact area of each microcontact is just a small proportion of the nominal contact area because of the imperfect contact between grooves and balls. But the microtopography of rough surfaces and the imperfect contact between the grooves and rolling elements was not considered in the above wear models of rolling components. Moreover, the contact load generated by the elastic, elastic-plastic, and plastic deformations of contacting asperities at the interface is equal to the contact load between balls and screw grooves. Namely, the microcontacts of all contacting asperities generate the total actual contact area and the contact load. But the multiscale contact characteristics of the contacting asperities in each contact elliptical area are not considered in the above models. So the multiscale contact mechanics model should be established to compute the above contact parameters and then it is essential to introduce the modified actual contact area into the precision loss model.

It is concluded that the precision loss modeling method of the ball screw pair is somewhat complex because of the imperfect contact between the grooves and balls. Moreover, the wear mechanism of the ball screw pair has not been fully revealed from the microscopic view, and the method to predict and reduce the wear of the screw groove has not been fully investigated so far. So it is essential to study the precision loss modeling method of screw groove. The general wear model is suitable to predict the wear between two rough surfaces, instead of the wear between two rough curved surfaces, which is the

case of the contact between grooves and balls. Then the general wear model fails in the accurate modeling of the precision loss of the rolling functional component. It is assumed that the grooves and balls are perfectly smooth in the special wear model of rolling functional components, which violates the law that any rough surface is a rough enough when viewed microscopically. The combination of the general wear model and the special wear model provides new solutions for the wear issue of the rolling functional components by considering the fractal behaviors of the grooves and balls. To predict the precision loss, the fractal characterization of rough surface has become the primary problem. By considering the effect the surface form on the contact parameters, the multiscale contact load model is established to realize the unification of macro- and micro- contacts. Finally, the precision loss model is proposed based on the multiscale contact load model with the aid of the kinematic parameters of the ball screw pair.

To improve the service life, the service performance, and the positioning accuracy of the screw nut pair, this paper focuses on a theoretical and experimental investigation into the precision loss modeling method by proposing a modified Archard wear model. Departing from former studies, the combined effect of intermittent contact between the balls and screw groove and the multiscale contact effect of surface microtopography is considered in the present work in such a way that it improves the modeling accuracy of the precision loss model and paves the way for improvement of positioning accuracy of BSFDSs. The main advantage of the precision loss model is its high modeling accuracy and the precision loss modeling of screw grooves is realized by unifying the characterization of the microtopography of rough surfaces with the multiscale contact mechanics model between two rough curved surfaces. The principle idea of the method is to introduce an effective contact coefficient into the Archard wear model to consider the intermittent contact between the balls and screw groove and then contact parameters in the Archard wear model are modified by characterizing the microtopography by the fractal theory and establishing the multiscale contact mechanics model between two rough curved surfaces. The multiscale contact mechanics model is established by considering the elastic, elastic-plastic, and plastic deformations of the contacting asperities in the wear process. Besides, a surface contact coefficient is introduced into the multiscale contact mechanics model to modify such contact parameters as the actual contact area and contact load by considering the contact between two rough curved surfaces. The framework of this paper is as follows: In Section 2, the microtopographies of the grooves and rolling elements were characterized by the fractal geometry theory. By considering the elastic, elastic-plastic, and plastic deformations of the contacting asperities on grooves and rolling elements, a multiscale contact mechanics model was established based on the characterization of the microtopography of the grooves and rolling elements to calculate such contact parameters as the total actual contact area and the contact load over the entire contact interface. Then, an effective surface contact coefficient was introduced into the multiscale contact mechanics model to correct the actual contact area and the contact load. Moreover, the sliding distance between a ball and the screw groove was computed and the nominal contact area is calculated by the Hertz contact theory. Finally, the model to predict the wear volume and precision loss of the screw groove was established based on the modified Archard theory. In Section 3, both the experimental setup and the measuring principle were introduced and then the wear experiments were conducted on a BSFDS under different feed rates and the effectiveness of the proposed model was verified by comparing and analyzing the predicted precision loss with experimental data of screw grooves. In Section 4, both the effects of the axial load and feed rates on the precision loss and the applicability and limitations of the proposed model were discussed.

## 2. Wear volume and precision loss modeling of screw groove

According to Archard wear theory [21], the wear volume between two rough surfaces is expressed as

$$V = K \cdot A_c \cdot L \quad (1)$$

where  $K$  denotes the wear coefficient;  $A_c$  denotes the actual contact area; and  $L$  denotes the relative sliding distance between two rough surfaces.

The contact between two rough surfaces of the rolling element and the screw groove is the contact between an equivalent rough and a rigid flat plane. In the forward and reverse strokes, all the rolling elements pass through the abraded areas 1 and 2, as shown in Fig. 1.  $L_S$  and  $L_R$  denote the effective travel range of the screw and the length of the ball screw groove, respectively. The relationship between them is written as  $L_R = L_S / \sin \alpha$ .

In the Archard theory, the sliding surfaces are in contact with each other all the time. However, only part of the screw groove in the nut contacts with the balls in the ball screw pair. As shown in Fig. 1, there are two abraded areas in the screw raceway. Balls are in contacts with the abraded area 1 and 2 during the forward stroke and reverse stroke, respectively. The shaded parts in the abraded areas, corresponding to the maximum Hertz's contact stress area of the contact ellipse, are the major wear regions of the screw groove. For the contact between the screw grooves and balls, the balls contact with screw grooves intermittently and the contact ellipses in the abraded areas actually participate in the wear process. Then the effective contact coefficient  $E_c$  is established as the ratio of the wear width to the total moving distance of balls to reflect the partial contact condition in ball screws. The overlay area between the contact ellipse and the major wear region is regarded as a rectangular with the width of  $2b_i$ . For the abraded area, the moving distance of the balls is the product of the number  $i_t$  of leads and the length  $L_R$  of the screw groove. There are two abraded areas, and then the total moving distance of the balls is  $i_t \cdot 2L_R$ . Finally, the effective contact coefficient is established as

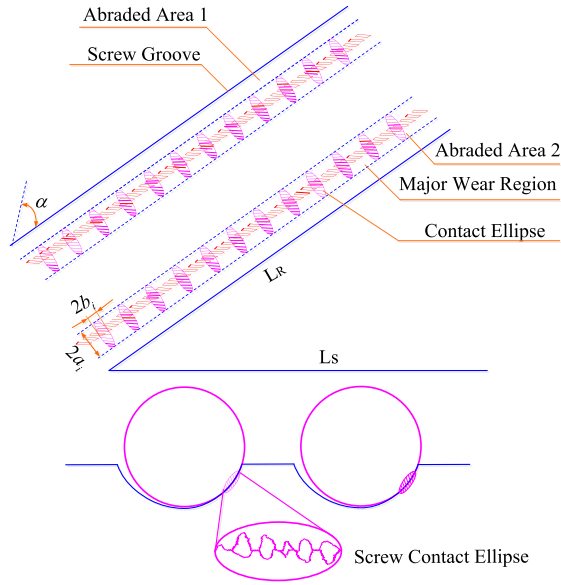


Fig. 1. Contact and wear mechanisms between the balls and the screw groove.

$$E_c = \frac{2b_i}{i_t \cdot 2L_R} = \frac{b_i \cdot \sin \alpha}{i_t \cdot L_S} \quad (2-a)$$

where  $\alpha$  and  $b_i$  denote the helix angle of the screw and the width of the major wear region, respectively; and  $i_t$  denotes the number of leads.

For the ball screw pair with the external circulation, the ball reenters the circulation process through the spiral groove or the cannula on the outer surface of the nut at the end of the circulation. For the ball screw pair with internal circulation, the reverser is used to realize the circulation of the balls. Namely, for the contact between the screw grooves and balls, the balls contact with screw grooves intermittently and the contact ellipses in the abraded areas actually participate in the wear process. So it is necessary to introduce the effective contact coefficient to modify the Archard wear model.

$$V = E_c \cdot K \cdot A_c \cdot L \quad (2-b)$$

To compute the wear volume, the parameters in Eq. (2b) should be determined. The actual contact area  $A_c$  is obtained by establishing the multiscale contact mechanics model and the sliding distance  $L$  is determined by conducting kinematic analysis of balls. In the following section, the method to determine the actual contact area  $A_c$  and the sliding distance  $L$  will be discussed.

### 2.1. Multiscale contact load model

The aim of this section is to determine the actual contact area  $A_c$  in the modified Archard wear model from the microscopic view. For each contact ellipse, the imperfect contact between each screw groove and ball leads to only a small portion of the contact ellipse participates the wear process, as shown in Fig. 1. The elastic, elastic-plastic and plastic deformations happen to the asperities with higher height and then the macro wear process happens because of the relative sliding between the screw groove and balls. When the external axial load is applied to the ball screw feed drive system, the elastic, elastic-plastic, and plastic deformations of the contacting asperities on the contact interface is generated to balance the external axial load. To get the actual contact area  $A_c$  in the Archard wear model, it is essential to analyze the deformation states of the contacting asperities. Then the total contact load  $Q_{total}$  and the actual contact area  $A_c$  at the screw groove/balls contact interface is obtained.

#### 2.1.1. Characterization of rough surface topography

The surface microtopography, which has an effect on the number of the contacting asperities, determines the wear volume between the screw groove and rolling element. So the characterization of the surface microtopography is the basis for the establishment of the precision loss model. Most traditional research methods on characterization of rough surface microtopography are based on statistical theory and the stochastic process theory. However, the characterization results obtained by these methods are sensitive to the sampling length and resolution of the measuring instrument. The surfaces of grooves and rolling elements are polished by mechanical machining, and then the surface microtopographies of the grooves and rolling elements are rough enough in the view of microscopic and show such intrinsic characteristics as disorder, random,

fractal, multiscale, and self-affinity features. The change of the surface microtopography is a nonstationary stochastic process with intrinsic fractal property, i.e., continuous everywhere but not differentiable and the above intrinsic behaviors of the microtopography of the grooves and rolling elements can be characterized by the fractal geometry. In the practical application, the two-dimensional (2D) Weierstrass-Mandelbrot [22] (W-M) function in fractal geometry is the most commonly used function to characterize the rough surface microtopography.

$$z(x) = G^{(D-1)} \sum_{n=n_1}^{\infty} \frac{\cos 2\pi \gamma^n x}{\gamma^{(2-D)n}}; \quad 1 < D < 2; \quad \gamma > 1 \quad (3)$$

where  $G$  denotes a scaling constant;  $D$  denotes the fractal dimension of the surface profile; the frequency expression denotes  $\omega = \gamma^n$  and  $\gamma$  is related to the sampling length  $L$ , by  $\gamma^{n_1} \approx 1/L$ , with  $n_1$  corresponding to the low cutoff frequency of the profile. The fractal roughness parameter  $D$ , varying from 1 to 2, is dimensionless, while the scaling parameter  $G$  is not limited to a specific range and has a length dimension.

### 2.1.2. Fractal parameter identification

The fractal surface is divided into isotropic and anisotropic surfaces. The isotropic surface means that any contour has the same statistical properties in any direction of the surface, and the anisotropic surface refers to different statistical properties on the surface in different directions. For the isotropic surface, the relationship between the 3D fractal dimension  $D_3$  and 2D fractal dimension  $D$  is expressed as

$$D_3 = D + 1 \quad (4)$$

where  $D_3$  denotes the 3D fractal dimension, and it is the fractal dimension of the surface. For the isotropic surface, the properties of the surface are obtained by studying the characteristics of its relatively simple profile. Moreover, the direction of the topographic data should be vertical to the texture direction induced by the mechanical machining.

For anisotropic surfaces, the 3D fractal dimension  $D_s$  of the surface is obtained if any two orthogonal contours are independent or weakly correlated.

$$D_s = D_{ver} + D_{hor} \quad (5)$$

where  $D_{ver}$  and  $D_{hor}$  denote dimensions of mutually orthogonal profiles of surfaces.

It is concluded that the rough surfaces are represented by 2D profiles and that it is a reasonable approximation for isotropic surfaces. For the anisotropic surfaces, the approximation is not valid anymore. According to the above analysis, the core for converting the measured three-dimensional topography data into two-dimensional topography data becomes as the determination of whether the rough surface is isotropic or anisotropic. The mechanical surface such as polishing, grinding, shot peening, etc. is generally isotropic. So the 2D fractal dimension  $D$  of the profile can be directly obtained according to the power spectrum method. Then Eq. (4) is used to get the 3D fractal dimension  $D_3$ . The surfaces machined by the finishing, the honing, and the flat grinding, etc. are anisotropic. Then Eq. (5) is used to determine the fractal dimension  $D_s$ .

According to Ref. [23], it is known that isotropic surfaces possess axially symmetric power spectra [24]. Then the topography data shown in Fig. 2(a) are used to calculate the power spectrum, as shown in Fig. 3. In the calculation, the fast Fourier transform is used. It is shown that the power spectrum shown in Fig. 3 exhibits axial symmetry, which means that the surface of the screw groove is an isotropic surface. Then the rough surfaces are represented by 2D profiles and it is a reasonable approximation. Moreover, the direction of the 2D profile is vertical to the texture direction induced by mechanical machining. Then the data in the X-direction, shown in Fig. 2(b), are used to calculate the 2D fractal dimension  $D$ . Moreover, the 3D fractal dimension  $D_3$  is then obtained according to Eq. (4). So the standard for converting the measured three-dimensional topography data into two-dimensional topographic data of the XOZ plane is that the measured surface is isotropic.

The fractal parameters are determined from a log-log plot of the power spectrum function of the W-M function versus wavelength [25–27]. The power spectrum of the W-M function is expressed as

$$S(\omega) = \frac{G^{2(D-1)}}{2\ln\gamma} \frac{1}{\omega^{(5-2D)}} \quad (6)$$

Then the power spectrum is taken logarithm on both sides and then Eq. (6) is expressed as

$$\lg S(\omega) = 2(D-1)\lg G - \lg 2\ln\gamma + (2D-5)\lg\omega \quad (7-a)$$

Then Eq. (7-a) is written as

$$\lg S(\omega) = k\lg\omega + b \quad (7-b)$$

where  $k = 2D - 5$ , and  $b = 2(D-1)\lg G - \lg 2\ln\gamma$ .

It is seen that the power spectrum function is a straight line in the double logarithmic coordinates and then the fractal parameters are identified as

$$\left\{ \begin{array}{l} D = \frac{(k+5)}{2} \\ \lg G = \frac{b+\lg(2\ln\gamma)}{2(D-1)} \end{array} \right\} \quad (8)$$

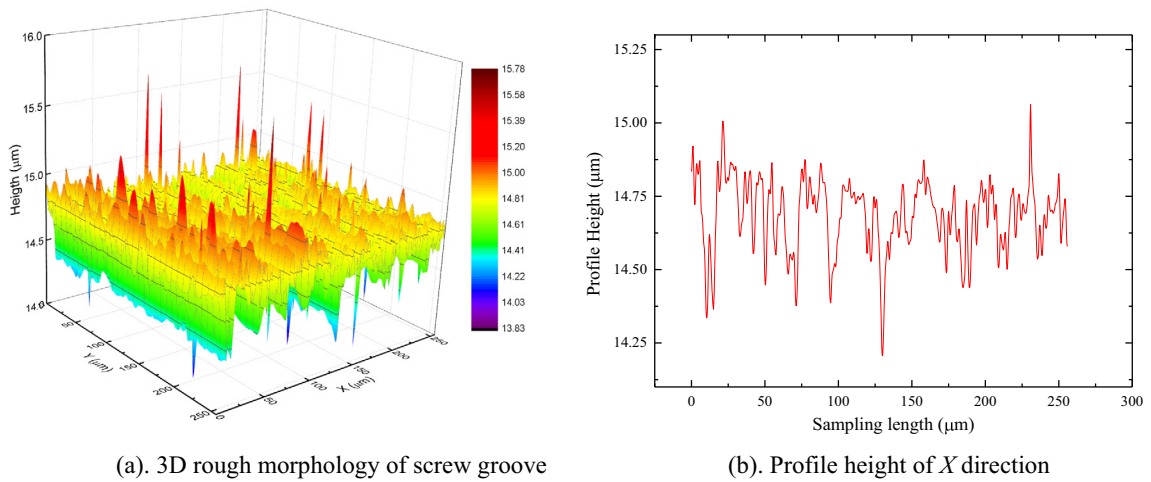


Fig. 2. Micro-morphology of screw shaft.

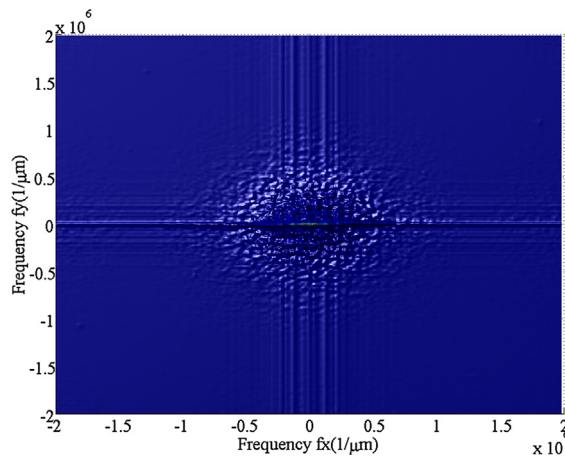


Fig. 3. Power spectrum of micro-morphology.

To identify the fractal parameters, the 3D surface microtopography of the screw groove was measured by an Olympus laser microscope LEXT OLS4000, as shown in Fig. 2. It can be seen that the grooves along the grinding direction are very obvious. The profile height along the X direction is shown in Fig. 2(b) and is regarded as the data source to identify the fractal parameters  $G$  and  $D$ .

The structure function of the profile height in the X direction is calculated by combining the data shown in Fig. 2(b) and Eq. (7-a), as shown in Fig. 4. As is shown, the power spectrum is close to a linear function. The least square fitting is used to obtain a straight line, and then the fractal parameters can be calculated according to Eq. (8). The identified fractal parameters are  $D = 1.62$  and  $G = 2.8e^{-11}$  m.

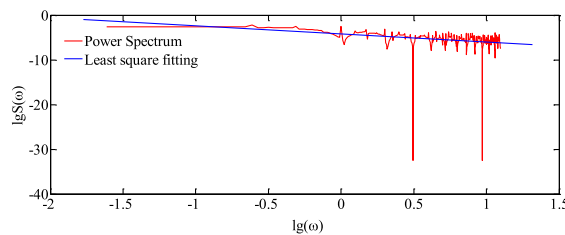


Fig. 4. The power spectrum of X direction morphology.



### 2.1.3. Multiscale contact mechanics model

To compute the contact parameters, it is essential to establish a multiscale contact mechanics model of the contact interface of groove/ball. It is assumed in the Hertz elastic contact theory that the contact of two bodies is the contact of two contacting asperities. However, rough surfaces are composed of a large number of asperities with different scales. So the Hertz elastic contact theory cannot fully reflect the true contact state of contacting bodies. The contact parameters at the contact interface of the screw grooves/rolling elements are determined by the deformation states of the contacting asperities with different scales on the rough surfaces of screw grooves and rolling elements. Namely, the elastic, elastic-plastic, and plastic deformations of asperities should be considered in the multiscale contact mechanics model.

**2.1.3.1. Surface contact coefficient.** The contact between the screw groove and balls is the contact between two rough curved surfaces, resulting in the changes in the contact load and the actual contact area. The contact between two rough surfaces happens at finite discrete asperities with different scales. Then the number of the asperities whose contact area is greater than  $a$  is expressed as

$$N(A > a) = \left(\frac{a_L}{a}\right)^{D/2} \quad (9)$$

where  $a_L$  denotes the maximum contact area; and  $a$  denotes the contact area of the asperity.

The area distribution of the contact points satisfies the power-law behavior. Then, the number of the asperities whose area is between  $a$  and  $a + da$  is obtained according to Eq. (9).

$$n(a) = -\frac{dN}{da} = \frac{D}{2} \frac{a_L^{D/2}}{a^{(D/2)+1}} \quad (10)$$

However, Eqs. (9) and (10) only apply to the contact between two rough surfaces. The contact between the screw groove and the rolling element is the contact between two rough curved surfaces. So Eq. (10) cannot be used to compute the number of the asperities whose area is between  $a$  and  $a + da$ . Eq. (10) is applicable when the surfaces are not curved, cylindrical, and spherical surfaces. However, the surfaces of grooves and balls are the spherical surfaces, and then the contact between them is the internal contact between two spherical surfaces. The change of the surface forms leads to the change in the fractal contact area, as shown in Fig. 5. Correspondingly, the actual total contact area, the contact load, and the wear volume will also change. It is seen that the total contact area decreases when the surface form is the spherical surface, which means that Eq. (10) is not suitable to compute the contact parameters between spherical surfaces. Instead, it only applies to the calculation of the contact parameters of the contact between two rough surfaces without curvatures.

Here the surface contact coefficient  $\lambda$  is introduced as

$$\lambda = \left[ \frac{\left( \frac{4}{\pi E'} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi(R_1 - r_b)} \right] \left( \frac{1}{r_b} - \frac{1}{R_1} \right) \quad (11)$$

where  $R_1$  and  $r_b$  denote the curvature radii of the screw grooves and the rolling elements, respectively; and  $E'$  denotes the equivalent elastic modulus and  $1/E' = (1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2$ , wherein  $\nu_1$ ,  $\nu_2$ ,  $E_1$ ,  $E_2$  denote Poisson's ratio and Young's elastic modulus of the two contacting materials, respectively.

For the contact between two rough curved surfaces, the number  $n'(a)$  of the asperities whose area is between  $a$  and  $a + da$  is modified as

$$n'(a) = \lambda \cdot n(a) = \left[ \frac{\left( \frac{4}{\pi E'} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi(R_1 - r_b)} \right] \left( \frac{1}{r_b} - \frac{1}{R_1} \right) \cdot n(a) \quad (12)$$

**2.1.3.2. Deformation properties of contact interface.** The contact point with the qualitative size  $l$ , the asperity top radius  $R$ , and the roughness amplitude  $\delta$  is shown in Fig. 6. If the detail in a smaller scale is ignored and the first term of the W-M function is taken, then the profile of rough surfaces is expressed as

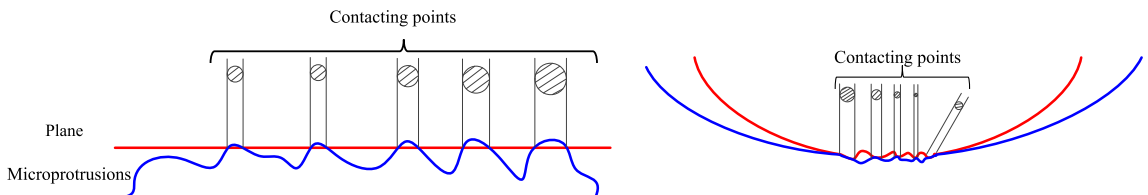


Fig. 5. Contact between two cylindrical surfaces.

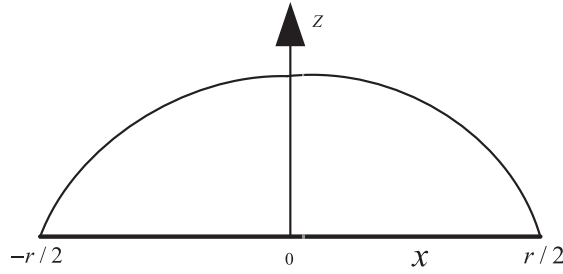


Fig. 6. Geometry of a contact spot of length scale  $r$ .

$$z(x) = G^{D-1} l^{2-D} \cos \frac{\pi x}{l} \quad (l/2 < x < l/2) \quad (13)$$

The top radius of the asperity is obtained according to Eq. (14).

$$R = \left| 1 / \left| \frac{d^2 z}{dx^2} \right|_{x=0} \right| = \frac{a^{D/2}}{\pi^2 G^{D-1}} \quad (14)$$

The correlation between the roughness amplitude  $\delta$  and the qualitative size  $l$  is expressed as

$$\delta = G^{D-1} l^{2-D} = G^{D-1} a^{(2-D)/2} \quad (15)$$

The critical elastic deformation  $\delta_{ec}$  and critical plastic deformation  $\delta_{pc}$  are crucial for the determination of the deformation states of the contacting asperities. So the critical elastic deformation  $\delta_{ec}$  and critical plastic deformation  $\delta_{pc}$  will be clarified first in the multiscale contact mechanics modeling.

#### 1) Elastic contact

The critical elastic deformation  $\delta_{ec}$  is obtained when the elastic deformation starts to happen.

$$\delta_{ec} = \left( \frac{3\pi p_m}{4E} \right)^2 \cdot R \quad (16)$$

where  $p_m$  denotes the average load applied to the contact.

Substituting Eq. (14) into Eq. (16), the following relationship is obtained.

$$\delta_{ec} = \left( \frac{3\pi p_m}{4E} \right)^2 \cdot \frac{a^{D/2}}{\pi^2 G^{(D-1)}} \quad (17)$$

The initial yield point occurs when  $p_m = 1.1Y = 1.1\sigma_y$ . Namely  $p_m = k\sigma_y$  and  $k = 1.1$ . When the asperity is under the fully plastic deformation,  $k = 3$ . Then

$$\delta_{ec} = \left( \frac{3k\sigma_y}{4E} \right)^2 \cdot \frac{a^{D/2}}{\pi^2 G^{(D-1)}} \quad (18)$$

When  $\delta < \delta_{ec}$  ( $a > a_{ec}$ ), the asperity is under elastic deformation and the critical elastic contact area  $a_{ec}$  is expressed as

$$a_{ec} = G^2 \left( \frac{3E^*}{4\pi k\sigma_s} \right)^{\frac{1}{D-1}} \quad (19)$$

#### 2) Elastic-plastic contact

The critical elastic-plastic contact area is determined by

$$\frac{E \cdot r}{\sigma_f \cdot R} = 30 \quad (20)$$

where  $r$  denotes the radius of the contact point and  $r = (a/\pi)^{1/2}$ ;  $\sigma_f$  denotes the plastic flow stress. For the ideal elastic-plastic material,  $\sigma_f = \sigma_s$ . For the strain hardening material,  $\sigma_f > \sigma_s$ . The relationship between  $\sigma_f$  and  $\sigma_s$  is expressed as

$$\sigma_f = \sigma_s \left( \frac{E \cdot \varepsilon}{\sigma_s} \right)^{1/m} \quad (21)$$



where  $\varepsilon$  denotes the total strain and can be expressed as  $\varepsilon = \sigma_s/E$ ; and  $m$  denotes the strain hardening index.

The strain of the asperity top is expressed as

$$\varepsilon = 0.2r/R \quad (22)$$

Substituting Eq. (22) into Eq. (21), the following relationship is obtained.

$$\sigma_f = \sigma_s \left( \frac{0.2 \cdot E \cdot r}{R \cdot \sigma_s} \right)^{1/m} \quad (23)$$

Assuming that  $\varphi = \frac{Er}{R\sigma_s}$ , then

$$\sigma_f = \sigma_s (0.2\varphi)^{1/m} \quad (24)$$

The relationship between the elastic-plastic deformation and contact area is obtained as

$$a_{ep} = \left( \frac{2m+1}{2m} \right)^{2(m-1)} \pi R \delta \quad (25)$$

By substituting Eq. (15) into Eq. (25), the relationship between the asperity top radius  $R_{ep}$  and the contact area  $a$  is obtained for the elastic-plastic deformation.

$$R_{ep} = \left( \frac{2m+1}{2m} \right)^{-2(m-1)} \frac{a^{D/2}}{\pi G^{(D-1)}} \quad (26)$$

Then the critical plastic deformation is expressed as

$$a_{pc} = G^2 \left( \frac{\beta E}{\sigma_s} \right)^{\frac{2}{D-1}} \quad (27)$$

where  $\beta = \frac{\left( \frac{2m+1}{2m} \right)^{-2(m-1)} \pi^{1/2}}{(30 \times 0.2^{1/m})^{\frac{m}{m-1}}}$

### 3) Plastic contact

When the contact area is less than the critical plastic transformation area  $a_{pc}$ , namely,  $a < a_{pc}$ , the pure plastic deformation happens to the contact points. Then the contact area is expressed as

$$a_p = 2\pi R_p \delta \quad (28)$$

Substituting Eq. (15) into Eq. (28), then the asperity top radius  $R_p$  for the elastic-plastic deformation is obtained.

$$R_p = \frac{a^{D/2}}{2\pi G^{(D-1)}} \quad (29)$$

When the contact area  $a$  is less than the critical plastic transformation area  $a_{pc}$ , namely,  $a < a_{pc}$ , the pure plastic deformation happens to the contact points. When the contact area  $a$  is less than the critical elastic transformation area  $a_{ec}$  and greater than the plastic transformation area  $a_{pc}$ , namely,  $a_{pc} < a < a_{ec}$ , the elastic-plastic deformation happens to the contact points. When the contact area  $a$  is greater than the critical elastic transformation area  $a_{ec}$ , the elastic deformation happens to the contact points.

**2.1.3.3. Contact load over entire contact interface.** The contact load on the contact interface of the screw groove/rolling element is the sum of the contact loads of all the individual contact points in each contact ellipse. When the contact area  $a$  is greater than the critical elastic contact area  $a_{ec}$ , namely  $a > a_{ec}$ , then elastic deformation happens to the contact point and the contact load is expressed as

$$P_e = \frac{4}{3\sqrt{\pi}} E G^{(D-1)} a^{(3-D)/2} \quad (30)$$

When the contact area  $a$  of the contact point is less than the critical elastic contact area  $a_{ec}$  and greater than the critical plastic contact area  $a_{pc}$ , namely  $a_{pc} < a < a_{ec}$ , the elastic-plastic deformation happens to the contact point and the contact load is expressed as

$$P_{ep} = \frac{2}{3} \sigma_f \left[ 2 + \ln \left( \frac{\varphi \sigma_s}{3\sigma_f} \right) \right] \quad (31)$$

Substituting Eq. (24) into Eq. (31), then

$$P_{ep} = p_{ep} \cdot a = \frac{2}{3} \sigma_s^{(m-1)/m} \cdot \left[ 0.2E^* \sqrt{\pi} \left( \frac{2m+1}{2m} \right)^{2(m-1)} \right]^{1/m} G^{(D-1)/m} \cdot a^{(2m+1-D)/2m} \cdot \left[ 2 + \ln \left( \frac{\varphi^{(1-\frac{1}{m})}}{3 \times 0.2^{1/m}} \right) \right] \quad (32)$$

When the contact area  $a$  of the contact point is less than the critical elastic contact area  $a_{pc}$ , namely  $a < a_{pc}$ , all contact points are under plastic deformation. Then, the contact load is expressed as

$$P_p = k\sigma_s a \quad (33)$$

Through the above analysis, the load on each asperity is obtained. By integrating the load on the contact interface from 0 to the maximum contact area  $a_L$ , the load distribution over each contact ellipse is obtained. When the fractal dimension  $D$  is not equal to 1.5, the contact load  $Q$  over the each contact ellipse is divided into three cases:

If  $a_L < a_{pc}$ , then

$$Q = k\sigma_s \int_0^{a_L} n'(a) da = k\sigma_s A_r \left[ \frac{\left( \frac{4}{\pi E} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi(R_1 - r_b)} \right]^{\left( \frac{1}{r_b} - \frac{1}{R_1} \right)} \quad (34-a)$$

If  $a_{pc} < a < a_{ec}$ , then

$$\begin{aligned} Q &= k\sigma_s \cdot \int_0^{a_{pc}} n'(a) da + \int_{a_{pc}}^{a_L} P_{ep} n'(a) da \\ &= \left[ \frac{\left( \frac{4}{\pi E} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi(R_1 - r_b)} \right]^{\left( \frac{1}{r_b} - \frac{1}{R_1} \right)} \left( k\sigma_s \frac{D}{2-D} a_L^{D/2} a_{pc}^{(2-D)/2} + \frac{k_3 \cdot k_5 - k_4}{(k_5)^2} (a_L^{k_5} - a_{pc}^{k_5}) + \frac{k_4}{k_5} (a_L^{k_5} \ln a_L - a_{pc}^{k_5} \ln a_{pc}) \right) \end{aligned} \quad (34-b)$$

where  $k_1 = \frac{2}{3} \sigma_s^{(m-1)/m} \cdot \left[ 0.2E^* \sqrt{\pi} \left( \frac{2m+1}{2m} \right)^{2(m-1)} \right]^{1/m}$ ;

$$k_2 = 2 - \ln(3 \times 0.2^{1/m}) + \frac{m-1}{m} \ln \left( \frac{E \sqrt{\pi} G^{(D-1)}}{\sigma_s} \cdot \left( \frac{2m+1}{2m} \right)^{2(m-1)} \right)$$

$$k_3 = \frac{D}{2} k_1 \cdot k_2 \cdot a_L^{D/2} \cdot G^{(D-1)/m}$$

$$k_4 = \frac{D}{2} k_1 \cdot a_L^{D/2} \cdot G^{(D-1)/m} \cdot \frac{(m-1)(1-D)}{2m}$$

$$k_5 = \frac{1+2m-D(1+m)}{2m}$$

If  $a > a_{ec}$ , then

$$\begin{aligned} Q &= k\sigma_s \int_0^{a_{pc}} n'(a) da + \int_{a_{pc}}^{a_{ec}} P_{ep} n'(a) da + \int_{a_{ec}}^{a_L} P_e n'(a) da \\ &= \left[ \frac{\left( \frac{4}{\pi E} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi(R_1 - r_b)} \right]^{\left( \frac{1}{r_b} - \frac{1}{R_1} \right)} \left[ k\sigma_s \frac{D}{2-D} a_L^{D/2} a_{pc}^{(2-D)/2} + \frac{k_3 \cdot k_5 - k_4}{(k_5)^2} (a_{ec}^{k_5} - a_{pc}^{k_5}) \right. \\ &\quad \left. + \frac{k_4}{k_5} (a_{ec}^{k_5} \ln a_{ec} - a_{pc}^{k_5} \ln a_{pc}) + \frac{k_4}{k_5} (a_{ec}^{k_5} \ln a_{ec} - a_{pc}^{k_5} \ln a_{pc}) + \frac{4D}{3\sqrt{\pi}(3-2D)} E^* a_L^{D/2} G^{(D-1)} (a_L^{(3-2D)/2} - a_{ec}^{(3-2D)/2}) \right] \end{aligned} \quad (34-c)$$

When the fractal dimension  $D$  is equal to 1.5, the load over the entire contact interface is given by

$$\begin{aligned} Q &= k\sigma_s \int_0^{a_{pc}} n'(a) da + \int_{a_{pc}}^{a_{ec}} P_{ep} n'(a) da + \int_{a_{ec}}^{a_L} P_e n'(a) da \\ &= \left[ \frac{\left( \frac{4}{\pi E} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi(R_1 - r_b)} \right]^{\left( \frac{1}{r_b} - \frac{1}{R_1} \right)} \left( 3k\sigma_s a_L^{3/4} a_{pc}^{1/4} + \frac{k_3 \cdot k_5 - k_4}{(k_5)^2} (a_{ec}^{k_5} - a_{pc}^{k_5}) + \frac{k_4}{k_5} (a_{ec}^{k_5} \ln a_{ec} - a_{pc}^{k_5} \ln a_{pc}) + \frac{E}{\sqrt{\pi}} a_L^{3/4} G^{1/2} (\ln a_L - \ln a_{ec}) \right) \end{aligned} \quad (34-d)$$

Then the contact load over the entire contact interface of rolling elements/screw grooves is expressed as

$$Q_{total} = Q \cdot i_t \cdot Z_C \cdot N_t \quad (34-e)$$

where  $N_t$  denote the number of the circles; and  $Z_C$  denotes the effective number of the balls in one circle.

**2.1.3.4. Total actual contact area over contact interface.** Through the above analysis, the contact area on each asperity is obtained, by integrating the contact area on the contact interface from 0 to the maximum contact area  $a_L$ , the contact area distribution over the entire contact interface is obtained. The actual contact area  $A_r$  in each contact ellipse is expressed as

$$A_r = \int_0^{a_L} n'(a) da = \left[ \frac{\left( \frac{4}{\pi E'} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi (R_1 - r_b)} \right]^{\left( \frac{1}{r_b} - \frac{1}{R_1} \right)} \cdot \frac{D}{2 - D} \cdot a_L \quad (35-a)$$

Then the total actual contact area  $A_c$  over the entire contact interface of ball/screw groove is expressed as

$$A_c = i_t \cdot Z_C \cdot N_t \cdot A_r = i_t \cdot Z_C \cdot N_t \cdot \left[ \frac{\left( \frac{4}{\pi E'} \frac{R_1 r_b}{R_1 - r_b} \right)^{1/2}}{\pi (R_1 - r_b)} \right]^{\left( \frac{1}{r_b} - \frac{1}{R_1} \right)} \cdot \frac{D}{2 - D} \cdot a_L \quad (35-b)$$

As is shown, the actual contact area  $A_c$  over the entire contact interface of the screw groove/rolling element is the function with the equivalent elastic modulus  $E$ , the fractal dimension  $D$ , the maximum contact area  $a_L$ , the number  $N_t$  of the circles, the number  $i_t$  of leads, and the effective number  $Z_C$  of the balls in one circle as its independent variables. Then the equivalent elastic modulus  $E$  is determined by  $1/E = (1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2$ . The fractal dimension  $D$  is identified by the power spectrum method shown in Section 2.1.2. The number  $N_t$  of the circles, the number  $i_t$  of leads, and the effective number  $Z_C$  of the balls in one circle are known for the ball screw pair. So far the value of the maximum contact area  $a_L$  has not been identified.

To keep balance, the external axial load  $F_{ext}$  is equal to the axial load caused by the normal contact load  $Q$ , namely

$$Q \cdot i_t \cdot Z_C \cdot N_t \cdot \sin \alpha \cdot \cos \theta = F_{ext} \quad (36)$$

where  $\alpha$  denotes the contact angle between the screw groove and the rolling element; and  $\theta$  denotes the helix angle of the screw groove. Then, the maximum contact area  $a_L$  is identified.

## 2.2. Relative sliding distance

The relative sliding distance between the rolling element and the screw groove is written as

$$L = |\vec{V}| \cdot t \quad (37)$$

where  $|\vec{V}|$  denotes the slipping velocity of the rolling elements with respect to the screw groove; and  $t$  denotes the running time.

The movement and contact parameters between the rolling element and the screw groove are shown in Fig. 7. The Frenet coordinate system  $(t, n, b)$  is used to characterize the moving path of the rolling elements. The angular velocity of the balls

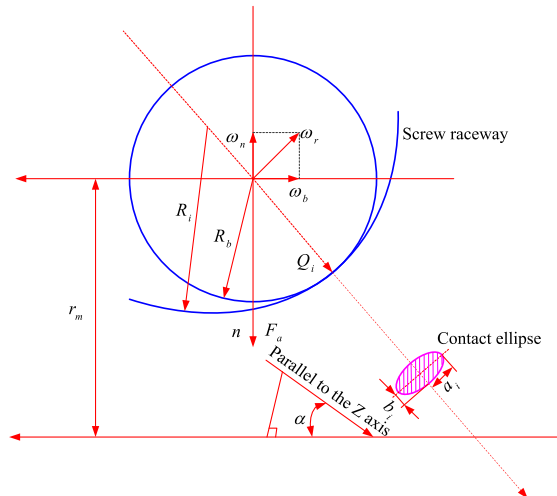


Fig. 7. Velocity components at the X and Y directions

with respect to the screw shaft is  $\omega_i$  and it is parallel to the  $b$ -direction. In an elliptical contact area, the relative slipping velocity  $|\vec{V}|$  between two surfaces is obtained by its velocity components in X- and Y- directions, respectively. Then the slipping velocity  $|\vec{V}|$  of the rolling elements is expressed as

$$|\vec{V}| = \sqrt{r_b^2(\omega_t - \omega \sin \alpha)^2 + \left\{ \frac{r_m}{\cos \alpha}(\omega_m - \omega) - r_b[(\omega_b - \omega \cos \alpha) \cos \alpha_i - \omega_n \sin \alpha_i] \right\}^2} \quad (38)$$

where  $r_m$  and  $\omega_m$  denote the radius of the screw and the angular velocity of the rolling elements, respectively;  $\omega$  denotes the rotational speed of the screw shaft; and  $\omega_t$ ,  $\omega_b$  and  $\omega_n$  denote the three absolute velocity components in the  $t$ -,  $b$ - and  $n$ -directions due to the spinning of the rolling element with an angular velocity of  $\omega_R$ , respectively.

### 2.3. Nominal contact area

The nominal contact area  $A_n$  of the balls/screw grooves is expressed as

$$A_n = \sum_{i=1}^{i_t \cdot Z_C \cdot N_t} \pi a_i b_i \quad (39-a)$$

where  $a_i$  and  $b_i$  denote the semimajor and semiminor of the contact ellipse, respectively.

According to the Hertz contact theory, the shape of the contact area is an ellipse under the effect of the normal contact load  $Q$  of the screw groove/rolling element. Then, the semimajor and semiminor are expressed as

$$\begin{cases} a_i = a^* \left( \frac{3Q}{E^* \Sigma \rho} \right)^{1/3} \\ b_i = b^* \left( \frac{3Q}{E^* \Sigma \rho} \right)^{1/3} \end{cases} \quad (39-b)$$

where  $a^*$  and  $b^*$  denote the contact coefficients of the ellipse semimajor and semiminor, respectively, and are expressed as

$$\begin{cases} a^* = \left( \frac{2k^2 \Gamma}{\pi} \right)^{1/3} \\ b^* = \left( \frac{2\Sigma}{\pi k} \right)^{1/3} \end{cases} \quad (39-c)$$

where  $\Gamma = \int_0^{\pi/2} \left[ 1 - \left( 1 - \frac{1}{k^2} \right) \sin^2 \phi \right]^{-1/2} d\phi$ ; and  $\Sigma = \int_0^{\pi/2} \left[ 1 - \left( 1 - \frac{1}{k^2} \right) \sin^2 \phi \right]^{1/2} d\phi$ .

Substituting Eq. (39-c) into Eq. (39-b), the following relationship is obtained

$$\begin{cases} a_i = \left( \frac{6k^2 \Sigma}{\pi E^* \Sigma \rho} \right)^{1/3} Q^{1/3} \\ b_i = \left( \frac{6\Sigma}{\pi k E^* \Sigma \rho} \right)^{1/3} Q^{1/3} \end{cases} \quad (39-d)$$

where  $(\Sigma \rho)_i = \frac{1}{D_w} \left( 4 - \frac{1}{f_i} + \frac{2\gamma_i}{1-\gamma_i} \right)$ .

### 2.4. Wear volume and precision loss model

According to the above analysis, the parameters are determined and the effective contact coefficient  $E_c$  is introduced into the Archard wear model. Then the wear volume is expressed as

$$V = E_c \cdot K \cdot A_c \cdot |\vec{V}| \cdot t \quad (40)$$

Then the wear depth on the contact interface between the rolling element and the screw groove is obtained according to the above analysis. Namely

$$\delta' = \frac{V}{A_n} = E_c \cdot K \cdot A_c \cdot \frac{|\vec{V}|}{A_n} \cdot t \quad (41)$$

Thus, the total wear depth  $\delta_{total}$  of the screw groove is expressed as

$$\delta_{total} = i_t Z_C N_t \delta' = i_t \cdot Z_C \cdot N_t \cdot E_c \cdot K \cdot A_c \cdot \frac{|\vec{V}|}{A_n} t = \delta_{Wt} \cdot t \quad (42)$$

where  $\delta_{Wt}$  denotes the precision loss rate of the screw groove, and it is expressed as

$$\delta_{Wt} = i_t \cdot Z_C \cdot N_t \cdot E_C \cdot K \cdot A_c \cdot \frac{|\vec{V}|}{A_n} \quad (43)$$

It is seen that the wear volume is a function of the effective contact coefficient  $E_C$ , the total actual contact area  $A_c$ , the number  $i_t$  of leads, the number  $N_t$  of the circles, the effective number  $Z_C$  of the balls in one circle, the wear coefficient  $K$ , the nominal contact area  $A_n$ , the slipping velocity  $|\vec{V}|$ , and the running time  $t$ . The number  $i_t$  of leads and the number  $N_t$  of the circles, the effective number  $Z_C$  of the balls in one circle and the wear coefficient  $K$  are obtained by the basic parameters of the ball screw pair. The effective contact coefficient  $E_C$  is obtained as  $E_C = \frac{2b_j}{i_t \cdot 2L_R} = \frac{b_j \cdot \sin \alpha}{i_t \cdot L_S}$ . The total actual contact area  $A_c$  is calculated by Eq. (35-b). The nominal contact area is  $A_n$  calculated by Eq. (39). According to the above analysis, the precision loss model of the screw groove is established. The model considers the actual contact area by establishing the multiscale contact mechanics model of the groove/ball contact interface and then the surface contact coefficient of the contact interface is introduced to modify such contact parameters as the actual contact area and contact load.

### 3. Precision loss modeling algorithms

The steps to solve the above precision loss model are as follows:

- (1) Input the basic parameters of the ball screw pair and the external axial load  $F_{ext}$ .
- (2) Compute the relative sliding distance  $L$  and the slipping velocity  $|\vec{V}|$  between two surfaces by Eqs. (37) and (38), respectively.
- (3) Compute the nominal contact area between the screw grooves and balls by Eq. (39).
- (4) Identify the fractal dimension  $D$  and scaling constant  $G$  by the power spectrum function method shown in Section 2.1.2.
- (5) Give the initial value for the maximum contact area  $a_L$  and then compute the total contact load  $Q_{total}$  by Eq. (34-e) and then compute the relative difference  $f$  between the total contact load  $Q_{total}$  and the  $F_{ext}$  by  $f = (Q_{total} - F_{ext})/F_{ext}$ . If  $|f| < \delta$ , then the multiscale mechanics model is convergent. If  $|f| > \delta$  and  $f > 0$ , then  $a_L = a_L + \Delta a_L$ . If  $|f| > \delta$  and  $f < 0$ , then  $a_L = a_L - \Delta a_L$ . Then the total contact load  $Q_{total}$  is computed by Eq. (34-e). The above process is repeated until the model is convergent. The criterion for setting the threshold  $\delta$  is as follows. The modeling accuracy is reduced if the threshold  $\delta$  is too big, and the calculation time will be reduced accordingly. The modeling accuracy is improved if the threshold  $\delta$  is too small, but the calculation time will increase dramatically. Then the threshold  $\delta$  is set as 0.001 according to the requirement of the modeling accuracy.
- (6) Calculate the actual contact area  $A_c$  by Eq. (35-b).
- (7) The wear volume and precision loss rate are calculated by Eqs. (40) and (43), respectively. The comprehensive algorithm is presented in Fig. 8.

### 4. Experiments and verification

#### 4.1. Experimental setup and measuring principle

To validate the effectiveness of the proposed precision modeling method, the wear experiments of the screw nut pair NSK W4009Z-E07ZSSTX-C2Z20 were conducted on a dual-drive ball-screw servo feed drive system under different axial loads and feed rates, as shown in Fig. 9. The basic parameters of the ball screw pair are listed in Table 1. The servo motors of the active and passive axis are controlled synchronously to ensure the normal and smooth travel of the worktable. The lubricant mode is grease. The maximum torque and the maximum feed rate of the ball screw feed drive system are 300 Nm and 64 m/min, respectively. The axial load  $F_a$  of the ball screw feed drive system is 2000 N. The ambient temperature is 21 °C in the whole wear test process.

The Renishaw laser interferometer XL80 was used to collect the precision loss of the ball screw groove. The measuring principle of the dual-drive ball screw servo feed system is shown in Fig. 10. The XL80 laser head is used to generate the laser and the linear interferometer and retroreflector are used to measure the precision loss. The air pressure and humidity sensors in the EC80 environmental compensation unit are used to compensate the fluctuations of the air pressure and humidity.

#### 4.2. Model validation

The experiment results could better represent the effects of axial load and feed rate on wear if different maximum feed rates or axial loads are utilized. Then three effective strokes are used in the experiments, and these three effective strokes correspond to three different feed rates. Namely, different maximum feed rates are used to in the experiments. The effective strokes of 300 mm, 600 mm, and 900 mm correspond to the feed rates of 100 mm/min, 200 mm/min, and 300 mm/min, respectively. The axial loads are the same in three experiments. The positioning error of the screw groove was measured every 125 h when the effective travel ranges of the nut are 300 mm, 600 mm and 900 mm by the Renishaw laser interferometer XL80, as shown in Fig. 11. The positioning error of the ball screw feed drive system increases with time for a specific

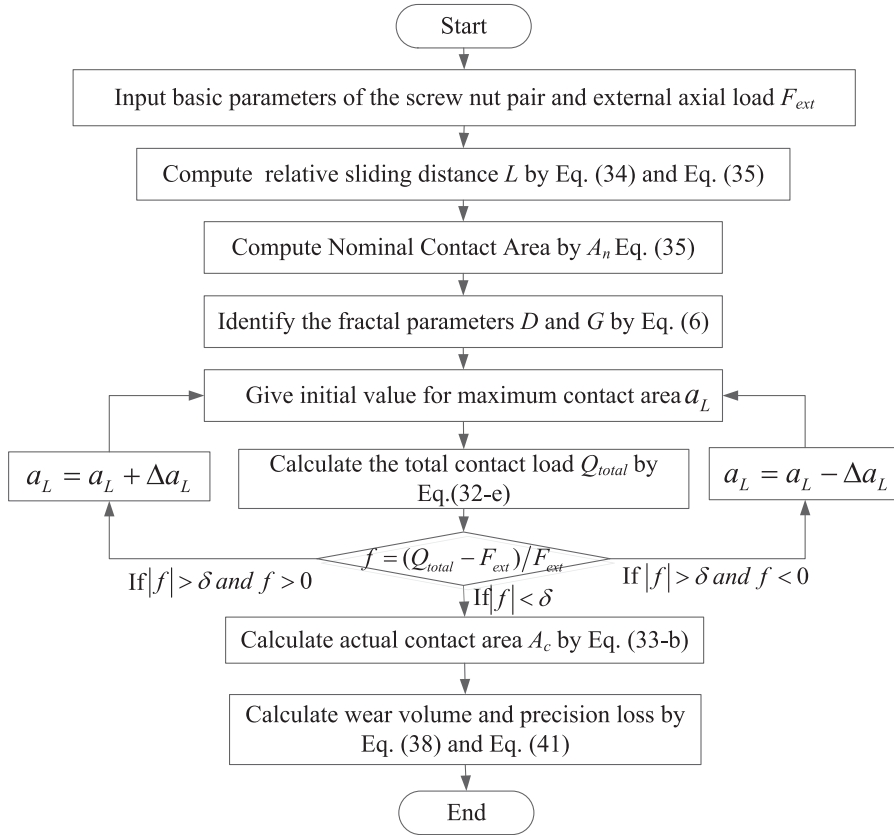


Fig. 8. Iterative algorithm for wear volume and precision loss modeling.

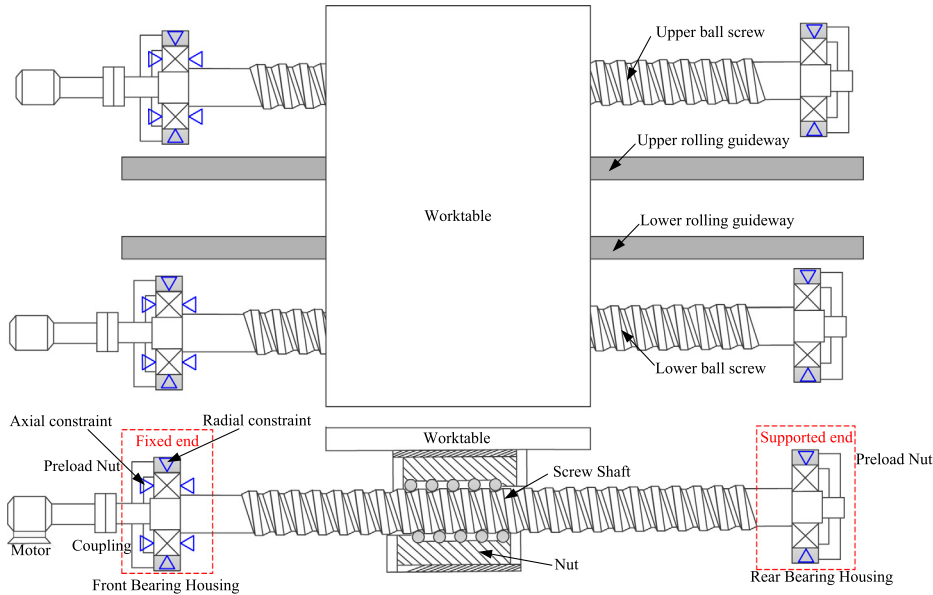


Fig. 9. Experimental setup.

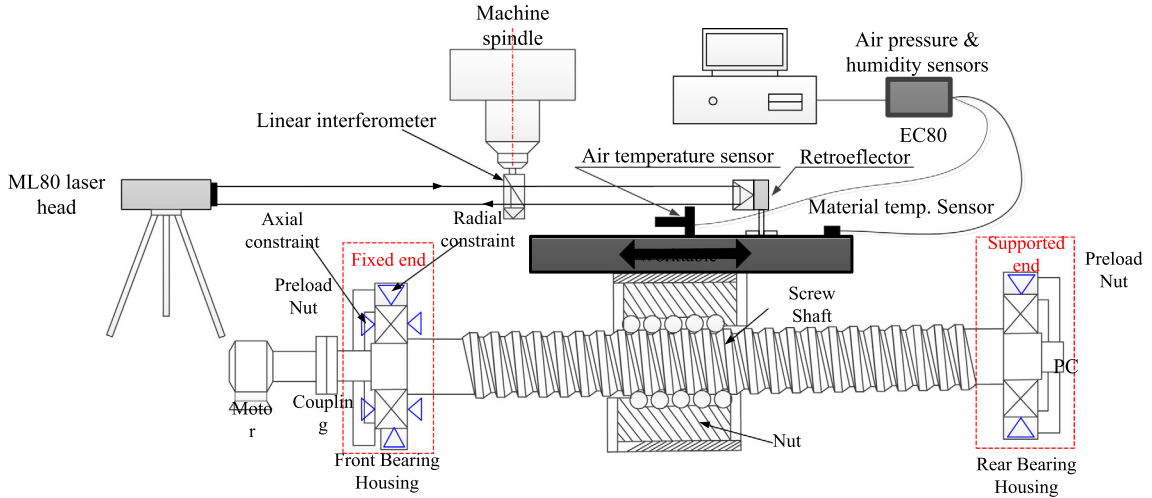
travel range. When the travel range is 900 mm, the positioning error of the dual-drive ball-screw feed drive system reduces from  $10.85 \mu\text{m}$  to  $8.68 \mu\text{m}$  in the first 125 h, which is known as a running-in process. Then, the positioning error of the dual-drive feed drive system increases from  $8.68 \mu\text{m}$  to  $10.98 \mu\text{m}$  from 125 h to 750 h because of the stable wear between the



**Table 1**

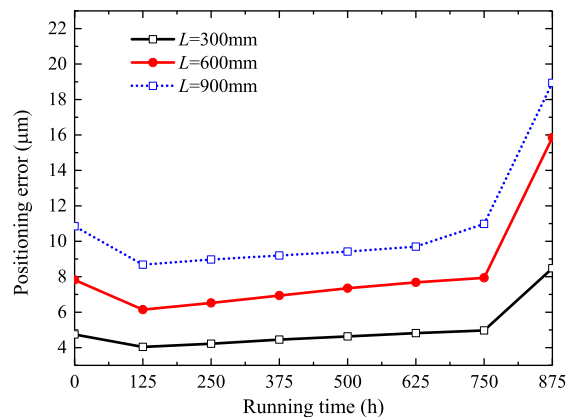
Parameters of ball-screw NSK W4009Z-E07ZSSTX-C2Z20.

| Parameters                  | Values   |
|-----------------------------|----------|
| Diameter of journal         | 40 mm    |
| lead                        | 20 mm    |
| Effective thread length     | 900 mm   |
| Contact angle               | 45°      |
| Helix angle                 | 8°12'    |
| Diameter of rolling element | 5.953 mm |
| Diameter of groove          | 6.43 mm  |

**Fig. 10.** Measuring principle of the dual-drive servo feed system.

screw groove and the rolling elements, which is the known as the stable wear stage. Finally, the positioning error of the dual-servo feed drive system increases dramatically from 10.98  $\mu\text{m}$  to 18.94  $\mu\text{m}$ , which is known as the excessive wear stage. Moreover, the positioning error at the same point increases with the travel range because the positioning error increases with the wear volume and the wear volume increases with the travel range. The precision loss increases dramatically after 750 h because the steady wear stage and the service life of the screw nut pair end up at 750 h.

To demonstrate the validity of the precision loss modeling method, the predicted precision loss obtained by the proposed model is compared with the experimental data and the results predicted by Wei model [10], as shown in Fig. 12. The predicted precision loss of the screw groove obtained by the proposed model agrees well with the measured data in the steady wear stage. In the excessive wear stage, both the present model and Wei model cannot predict the precision loss of the screw

**Fig. 11.** Measured positioning error of the ball screw feed drive system.

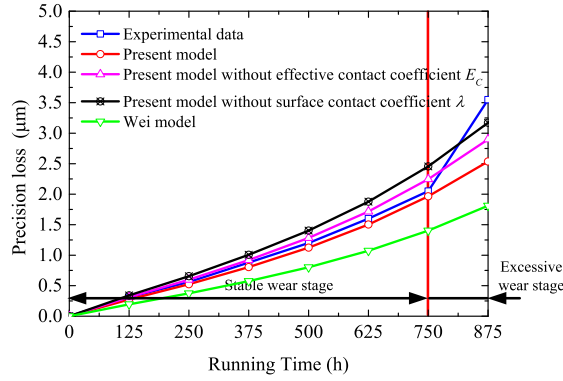


Fig. 12. Precision loss of the screw groove.

groove accurately because the present model is established for the stable wear stage and is not applicable to the wear issue in the excessive wear stage. The predictive values obtained by Wei model are relatively lower than the measured data in the whole service life and cannot reflect the dynamic change of the precision loss. An effective contact coefficient  $E_c$  is introduced into the Archard wear model to take the intermittent contact between the balls and screw grooves into account and then the multiscale contact mechanics model is established to consider the multiscale contact characteristics of the contacting asperities. Moreover, a surface contact coefficient  $\lambda$  is introduced into the multiscale contact mechanics model to modify the total actual contact area  $A_c$  used in the precision loss model, resulting in high predictive accuracy of the present model in the stable wear stage. The precision loss obtained by the present model without introducing the effective contact coefficient  $E_c$  is greater than that with introducing the effective contact coefficient  $E_c$  because the effective contact coefficient  $E_c < 1$ . The precision loss obtained by the present model without introducing the surface contact coefficient  $\lambda$  is greater than that with introducing the surface contact coefficient  $\lambda$  because  $\lambda < 1$ , as shown in Fig. 13. The surface contact coefficient  $\lambda$  is smaller than 1 in the whole range of the radius of balls. So  $n'(a)$  is smaller than  $n(a)$  for those whose contact areas are between  $a$  and  $a + da$  according to Eq. (10). The surface contact coefficient is smaller than 1 and it increases with the ball radius and then saturates to 1 when the ball radius is infinity, and then the multiscale contact mechanics model degenerates into the contact between two rough surfaces without curvatures.

The relative error between the experimental data and the predicted precision loss is expressed as

$$\eta = \frac{|S_{Exp} - S_{Pre}|}{S_{Exp}} \quad (44)$$

where  $S_{Exp}$  and  $S_{Pre}$  denote the experimental data and the predicted results, respectively.

The relative error between the experimental data and the predicted precision loss is obtained, as shown in Table 2. The relative errors in the stable wear stage are less than 11.94% and the relative error in the excessive wear stage is 28.56%. It is concluded that the predictive accuracy of the present model is much higher than that of the Wei model and that of without introducing the effective contact coefficient  $E_c$  and the surface contact coefficient  $\lambda$ . The maximum predictive error in the stable wear stage for Wei model is about 37.10% and the relative error in the excessive wear stage is 48.97%. For the present

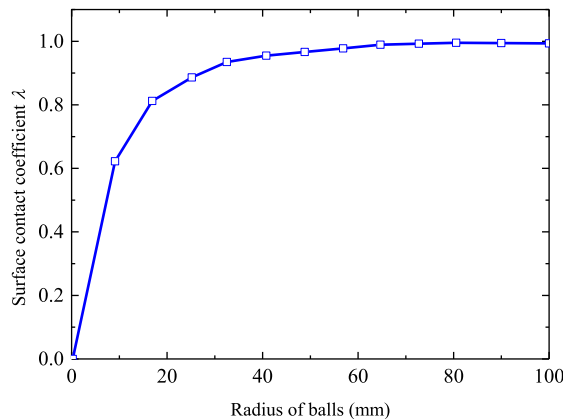


Fig. 13. Surface contact coefficient  $\lambda$ .

**Table 2**

Comparison between measured and predicted precision loss of the screw groove.

| Running time<br>(h) | Relative error between the experimental data and the predicted precision loss (%) |                                                           |                                                             |           |
|---------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|-----------|
|                     | Present model                                                                     | Present model without effective contact coefficient $E_c$ | Present model without surface contact coefficient $\lambda$ | Wei model |
| 0                   | 0                                                                                 | 0                                                         | 0                                                           | 0         |
| 125                 | 11.94                                                                             | 14.89                                                     | 12.35                                                       | 37.10     |
| 250                 | 6.38                                                                              | 16.82                                                     | 9.80                                                        | 33.12     |
| 375                 | 8.5                                                                               | 14.92                                                     | 12.31                                                       | 34.65     |
| 500                 | 6.41                                                                              | 14.64                                                     | 12.68                                                       | 33.15     |
| 625                 | 6.10                                                                              | 12.87                                                     | 15.01                                                       | 32.93     |
| 750                 | 4.16                                                                              | 13.05                                                     | 14.27                                                       | 31.54     |
| 875                 | 28.56                                                                             | 49.57                                                     | 33.44                                                       | 48.97     |

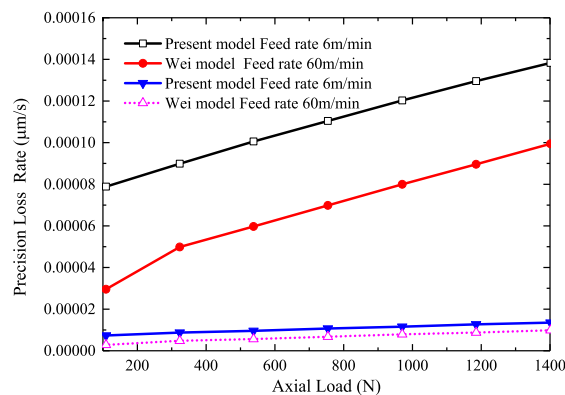
model without introducing the effective contact coefficient  $E_c$ , the maximum predictive error in the stable wear stage is 16.82%, and the relative error in the excessive wear stage is 49.57%. For the present model without introducing the surface contact coefficient  $\lambda$ , the maximum predictive error in the stable wear stage is 15.01% and the relative error in the excessive wear stage is 33.44%. The above analysis demonstrates that the introduction of the effective contact coefficient  $E_c$  is essential for the precision loss modeling and that the introduction of the surface contact coefficient  $\lambda$  is essential for the modification of the actual contact area used in the precision loss modeling.

## 5. Results and discussion

The axial load and the feed rate are the main factors which affect the precision loss, and then the precision loss rates are predicted by the present model under different working conditions to show the effect of the above factors on the precision loss rate of the screw groove.

### 5.1. Effect of axial load

The effect of the axial load on the precision loss rate is shown in Fig. 14 and the precision loss rate is almost proportional to the axial load for a specific feed rate. The total contact load  $Q_{total}$  on the ball-screw groove interface increases with the axial load  $F_{ext}$  according to Eq. (36), and then both the wear volume and precision loss of the screw groove increase with the total contact load  $Q$  according to Eqs. (40) and (43). Correspondingly, the maximum contact area  $a_L$  increases with the total contact load  $Q_{total}$  on the groove/ball contact interface according to Eq. (34). The increase in the maximum contact area  $a_L$  means that more contacting asperities participate in the wear process. The number of the asperities whose contact area is greater than  $a$  increases according to Eq. (9), which means the elliptical contact area and the contact load of each contact point increase with the axial load. According to the above analysis, the macro external load  $F_{ext}$  is passed to each contacting asperity and the increase in the axial load leads to the increase in the actual contact area  $A_r$  and contact load  $Q$  on each contact ellipse according to Eq. (35-a) and Eq. (34). Moreover, both the relative sliding distance  $L$  and the slipping velocity  $|\vec{V}|$  of the rolling elements increase with the feed rate according to Eqs. (37) and (38), and then the wear volume and precision loss rate increase with the feed rate according to Eqs. (40) and (43). Besides, the precision loss rate increases with the feed rate because the wear frequency between the screw groove and the rolling element increases with the feed rate.



**Fig. 14.** Comparison between the predicted accuracy loss rates by present model and Wei model.

## 5.2. Effect of feed rate

The predicted precision loss rate by the proposed model increases with the feed rate, as shown in Fig. 15. Moreover, the precision loss rates of the proposed model are not significant when the feed rate is 4 m/min, but the precision loss rates of the present model increase remarkably when the feed rate is 28 m/min. Therefore, it is essential to limit the feed rate to reduce the precision loss rate and extend the service time of the ball screw feed drive system. The precision loss rates increase with the feed rate at a given axial load because the increase in the feed rate leads to the increase in the slipping velocity  $|\vec{V}|$  and the relative sliding distance  $L$  according to Eqs. (37) and (38) and the precision loss rate increases the slipping velocity  $|\vec{V}|$  according to Eq. (43).

## 5.3. Applicability and limitations of the predictive model

### 5.3.1. Applicability of predictive model

The surface contact coefficient  $\lambda$  shown in Eq. (11) is applicable to the inner contact of two general curved surfaces. For the external contact of two general curved surfaces, the surface contact coefficient  $\lambda$  should be revised accordingly.

$$\lambda = \left[ \frac{\left( \frac{4}{\pi E} \frac{R_1 r_b}{R_1 + r_b} \right)^{1/2} \left( \frac{1}{r_b} + \frac{1}{R_1} \right)}{\pi (R_1 + r_b)} \right] \quad (45)$$

The surface contact coefficients for the external and inner contacts of two general curved surfaces are compared, as shown in Fig. 16. It is seen that both the surface contact coefficients for the external and inner contacts of two general curved surfaces are smaller than 1, which means that just a small portion of the rough surface participates in the contact and wear

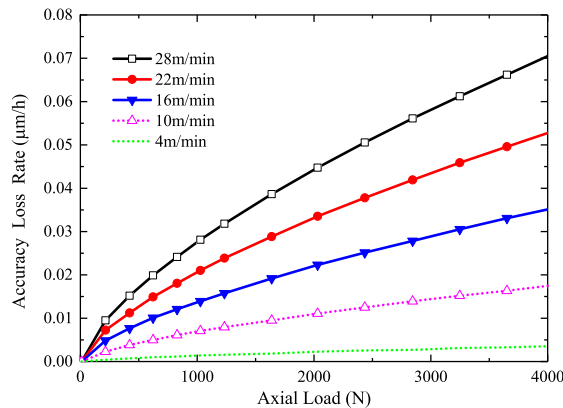


Fig. 15. Predicted accuracy loss rate of the screw raceway by proposed model under different axial load and rotational speed.

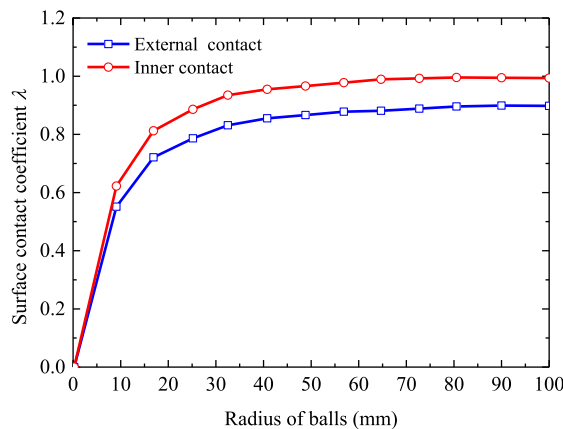


Fig. 16. Surface contact coefficient  $\lambda$  for the external and inner contacts of two general curved surfaces.

processes. For the inner contact between two rough curved surfaces, more contact area enters the actual contact when the contact changes from the external contact to inner contact. So the wear volume and precision loss for the external and inner contacts of two general curved surfaces will be different. Moreover, the proposed model is also applicable to predict the wear volume and precision loss rate for the contact interface of two rough surfaces by setting the surface contact coefficient  $\lambda$  and the effective contact coefficient  $E_c$  as 1, which is the case that the radii of balls and grooves are infinity. Besides, the model can be extended to predict the wear volume and precision loss rates of the bearing pair and the rolling linear guide pair. The multiscale contact load model can be extended to compute the actual contact area and contact load for the contact interface of two rough surfaces by setting the surface contact coefficient  $\lambda$  and the effective contact coefficient  $E_c$  as 1.

### 5.3.2. Limitations of predictive model

- (1) The inaccuracy of the predicted results is caused by the assumption that the microtopography of screw groove is strictly fractal. The assumption is valid when the surface of the screw groove is reasonably machined. However, the microtopography obtained by the mechanical machining is always non-strict fractal and the non-strict fractal characteristics can be found from the power spectrum of rough profile. The power spectrum is close to a linear function in higher-frequency range, while in lower-frequency range the error is relatively large, as shown in Fig. 4.
- (2) The surface contact coefficient introduced into the multiscale contact mechanics model is not applicable for the inner contact with identical radii according to Eq. (11). For the inner contact with identical radii, the method to construct the surface contact coefficient should be investigated further.
- (3) When the ball screw is actually working, the load direction and size are changed. The direction and size of the axial load applied on the ball screw pair are the same in the experiments. Then the wear behaviors of the ball screw pair under variable axial load cannot be studied. It is shown that the wear behaviors of the ball screw pair under variable axial loads are slightly different from that of the ball screw pair under constant axial load.

## 6. Conclusions

A precision loss model of the screw groove was established by considering the combined effect of the intermittent contact between the screw groove and balls and the multiscale effect of surface microtopography. The microtopographies of rough surfaces of screw grooves and balls were characterized by the fractal geometry theory. The multiscale contact mechanics model between two rough curved surfaces was established by considering the elastic, elastic-plastic, and plastic deformations of contacting asperities to compute the total actual contact area used in the precision loss model. The surface contact coefficient was introduced into the multiscale contact mechanics model to modify the contact load by considering the rough curved surfaces of balls and screw groove. Then the characterization of the microtopography of screw groove surfaces and the multiscale contact mechanics model of the groove/ball contact interface were unified in the modeling of the precision loss model. The sliding distance was calculated according to the kinematic analysis of the ball screw. Finally the wear volume and precision loss of the ball-screw contact were proposed by the above analysis. The predicted precision loss matches well with the experiment data in the stable wear stage and the following conclusions are obtained.

- (1) The introduction of the effective contact coefficient  $E_c$  into the precision loss model is essential because the intermittent contact between the balls and screw groove, which is helpful for the improvement of the predictive accuracy of the wear volume and precision loss.
- (2) To accurately predict the wear volume and the precision loss of rough curved surfaces, it is essential to introduce the surface contact coefficient  $\lambda$  into the predictive model of the precision loss to modify the actual contact area used in the precision loss modeling because the rough curved surfaces of balls and screw grooves change the actual contact area. The surface contact coefficient is smaller than 1 and it increases with the ball radius and then saturates to 1 when the ball radius is infinity and the multiscale contact mechanics model degenerates into contact between two rough surfaces.
- (3) The proposed multiscale contact mechanics model is effective to compute the total actual contact area and contact load at the contact interface. The contact pressure is generated by the elastic and plastic deformations of the contact points. The contact pressure at the contact interface of the groove/ball is balanced with the external axial load. Both the actual contact area and contact pressure increase with the axial load. The multiscale contact load model can be extended to compute the actual contact area and contact load for the contact interface of two rough surfaces by setting the surface contact coefficient  $\lambda$  and the effective contact coefficient  $E_c$  as 1.
- (4) The precision loss rate of the ball screw pair increases with the axial load because the increase in the axial load leads to the increase in the actual contact area by increasing the contact load. The precision loss rates increase with the feed rate at a given axial load because the increase in feed rate leads to the increase in the slipping velocity and precision loss rate increases with the slipping velocity.
- (5) It is effective to use the proposed precision loss model to predict the wear volume and the precision loss of the inner contact between two rough curved surfaces. The error between the measured data and the predicted precision loss in the stable wear stage is less than 11.94%. To predict the wear volume and precision loss between two general curved

surfaces, the surface contact coefficient should be revised according to the arrangements of the rough curved surfaces. Moreover, the model could be extended to the contact and wear between two general rough surfaces. This situation is a limit case of the model, namely,  $r_b \rightarrow \infty$  and the surface contact coefficient  $\lambda = 1$ .

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## Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work; there is no professional or other personal interest of any nature or kind in any product or company that could be construed as influencing the position presented in, or the review of, the manuscript.

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